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S.E. (Comp/IT.) (Second Semesters) EXAMINATION, 2017

ENGINEERING MATHEMATICS—III

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt *four* questions : Q. 1 or Q. 2; Q. 3 or Q. 4,
Q. 5 or Q. 6, Q. 7 or Q. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Use of non-programmable electronic pocket calculator is allowed.

(v) Assume suitable data, if necessary.

1. (a) Solve any two : [8]

(i) $(D^4 - 1)y = \cosh x \sinh x$

(ii) $(D^2 - 4D + 4)y = e^{2x} \sec^2 x$ (By variation of parameters)

(iii) $(x+1)^2 \frac{d^2 y}{dx^2} + (x+1) \frac{dy}{dx} = (2x+3)(2x+4)$

(b) Find the Fourier sine integral of : [4]

$f(x) = x^2, 0 < x < a$

$= 0, x > a$

P.T.O.

Or

2. (a) An electric circuit consists of an inductance 0.1 henry, a resistance R of 20 ohms and a condenser of capacitance C of 25 microfarads. If the differential equation of electric circuit is :

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0$$
, then find the charge q and current i at any time t , given that at $t = 0$, $q = 0.05$ Coulombs,

$$i = \frac{dq}{dt} = 0 \text{ when } t = 0.$$

- (b) Find the Inverse Z-transform (any one) :

(i) $F(z) = \frac{1}{(z-a)^3}$ (By using Inversion Integral Method).

(ii) $F(z) = \frac{z^2}{\left(z - \frac{1}{4}\right)\left(z - \frac{1}{5}\right)}$, $|z| > \frac{1}{4}$

- (c) Solve the following difference equation to find $\{f(k)\}$:

$$f(k+2) + 3f(k+1) + 2f(k) = 0,$$

$$f(0) = 0, f(1) = 1.$$

3. (a) Calculate the correlation coefficient for the following data :

x	1	2	3	4	5
y	2	5	2	7	6

- (b) A firm produces articles of which 0.1% are defective out of 600 articles. If wholesaler purchases 1000 such cases, how many can be expected to have two defectives ?

- (c) Find the angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 8$ at the point $(1, -2, 1)$. [4]

Or

4. (a) Find the directional derivative of $xz^3 - x^2yz$ at the point $(2, 1, -1)$ in the direction of tangent to the curve $x = e^t \cos t, y = e^t \sin t, z = e^t$ at $t = 0$. [4]
- (b) If $\vec{u}$ and $\vec{v}$ are irrotational vectors, then prove that $\vec{u} \times \vec{v}$ is solenoidal vector. [4]
- (c) A random sample of 500 screws is drawn from a population which represents the size of screws. If a sample is distributed normally with a mean 3.15 cm and standard deviation 0.025 cm, find expected number of screws whose size falls between 3.12 cm and 3.2 cm. (Given for $z = 1.2$, area = 0.3849, $z = 2.0$, area = 0.4772). [4]
5. (a) Evaluate :

$$\int_C \vec{F} \cdot d\vec{r} \text{ where } \vec{F} = z\vec{i} + x\vec{j} + y\vec{k} \text{ and}$$

C is the arc of the curve $x = \cos t, y = \sin t, z = t$ from $t = 0$ to $t = \pi$ [5]

- (b) Evaluate $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$ for $\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$ where S is the surface of paraboloid $z = 9 - x^2 - y^2, z \geq 0$. [4]

- (c) If $\vec{E} = \nabla\phi$ and $\nabla^2\phi = -4\pi\rho$, then prove that $\iint_S \vec{E} \cdot d\vec{S} = -4\pi \iiint_V \rho \, dv$. [4]

Or

6. (a) Using Green's theorem, evaluate :

$\int_C \left(\frac{1}{y} dx + \frac{1}{x} dy \right)$ where C is the boundary of the region bounded by the parabola $y = \sqrt{x}$ and line $x = 1$ and $x = 4$. [5]

- (b) Use divergence theorem to evaluate

$$\iiint_S (y^2 z^2 \bar{i} + z^2 x^2 \bar{j} + x^2 y^2 \bar{k}) \cdot d\bar{S}$$

where S is the upper part of the sphere $x^2 + y^2 + z^2 = 9$ above XOY plane. [4]

- (c) Prove that :

$$\int_C (\bar{a} \times \bar{r}) \cdot d\bar{r} = 2\bar{a} \cdot \iint_S d\bar{S}$$

where S is any open surface with boundary C. [4]

7. (a) Determine the analytic function $f(z) = u + iv$ in terms of z . Whose real part is $e^{2x}(x \cos 2y - y \sin 2y)$. [5]

- (b) Using Cauchy's Integral Formula evaluate $\oint_C \frac{\cos \pi z}{z^2 - 1} dz$ where C is the rectangle with vertices $2 \pm i, -2 \pm i$. [4]

- (c) Find the bilinear transformation which maps the points 1, i , -1 from z -plane onto the points i , 0, $-i$ of the W -plane. [4]

Or

8. (a) If $f(z) = u + iv$ be an analytic function find $f(z)$. If $u + v = r^2 (\cos 2\theta + \sin 2\theta)$. [5]

- (b) Using residue theorem evaluate : [4]

$$\int_C \frac{z^3 - 5}{(z+1)^2(z-2)} dz \text{ where } C \text{ is } |z| = \frac{3}{2}.$$

- (c) Find the mapping of the line $2y = x$ under the transformation

$$W = \frac{2z-1}{2z+1}. \quad [4]$$